Vilnius University Šiauliai Academy

Object Oriented Programming
Practical Work 1:
3x+1 problem

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Testing Report

Program description

The purpose of the program is to implement the Collatz conjecture successfully.

Test 1: Program initialization

```
 --Welcome, here is a practical demonstration of the Collatz sequence!--
Enter a number to receive Collatz (number must be >= 1):
```

Result: Upon running the program in the terminal, it successfully initialized.

Test 2: Input validation using a letter, negative number, zero, and a special character

```
Enter a number to receive Collatz (number must be >= 1): b
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1): ]
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1): -1
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1): 0
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1):
```

Result: In each case the program successfully returned the user to the original prompt.

Test 3: Calculation accuracy

```
Enter a number to receive Collatz (number must be >= 1): 5
Thank you. Here is the Collatz sequence!
----------------------------------------
Sequence step 1: 5
Sequence step 2: 16
Sequence step 3: 8
Sequence step 4: 4
Sequence step 5: 2
Sequence step 6: 1
----------------------------------------
```

Result: Upon inputting a correct number the program successfully ran.

Test 4: Program continuation input validation

-The user is given the choice whether to continue running the program, or not. Inputs besides Y/N were given:

```
----------------------------------------
Do you wish to continue? Y/N: b
Y FOR YES. OR N FOR NO.
----------------------------------------
Do you wish to continue? Y/N: f
Y FOR YES. OR N FOR NO.
----------------------------------------
Do you wish to continue? Y/N: -5
Y FOR YES. OR N FOR NO.
----------------------------------------
Do you wish to continue? Y/N: !
Y FOR YES. OR N FOR NO.
----------------------------------------
Do you wish to continue? Y/N: .
Y FOR YES. OR N FOR NO.
----------------------------------------
Do you wish to continue? Y/N: 5
Y FOR YES. OR N FOR NO.
----------------------------------------
```

Result: In each circumstance, the input validation ran successfully.

Test 5: Program continuation

```
----------------------------------------
Do you wish to continue? Y/N: Y
Enter a number to receive Collatz (number must be >= 1):
```

```
----------------------------------------
Do you wish to continue? Y/N: y
Enter a number to receive Collatz (number must be >= 1):
```

Result: If the user correctly enters Y, or alternatively y, the program restarts successfully.

Test 6: Program termination

```
----------------------------------------
Do you wish to continue? Y/N: N
Alright. Geros Dienos.
```

```
----------------------------------------
Do you wish to continue? Y/N: n
Alright. Geros Dienos.
```

Result: Upon receiving the answer, if the user responds with N or n, the program successfully terminates.

Conclusion

All the tests conducted were passed successfully. The program outputted correct Collatz sequences for the numbers provided. The program correctly implements input validation, supports repeated execution, and successfully terminates.

References:

Microsoft. (n.d.). Visual Studio Code [Computer Software]. From https://code.visualstudio.com